

Robust Precoding Techniques for Multibeam Mobile Satellite Systems

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Abstract— This paper presents designing precoding technique at the gateway of a multibeam mobile satellite systems, enabling full frequency reuse pattern among the beams. Such a system brings in two critical challenges to overcome. The inter-beam interference makes applying interference mitigation techniques necessary. Further, when the user terminals are mobile the Channel State Information (CSI) becomes time-varying which is another challenge to overcome. Therefore, the gateway has only access to an outdated CSI, which can eventually limit the precoding gains. In this way, employing a proper CSI estimation mechanism at the gateway can improve the performance of the precoding scheme. In this context, the objectives of this paper are two folds. First, we present different CSI feedback mechanisms which aim at preserving a lower CSI variations at the gateway. Then, we develop the corresponding precoding schemes which are adapted with the proposed CSI feedback mechanisms. To keep the complexity of the proposed precoding schemes affordable, we consider a maritime communication scenario so that the signals received by mobile user terminals suffer from a lower pathloss compared to the Land Mobile communication. Finally, we provide several simulations results in order to evaluate the performance of the proposed precoding techniques.

Index Terms—MIMO satellite, Mobile satellite systems, multi-beam satellite systems, Precoding schemes.

I. INTRODUCTION

A. Motivation

The ever-growing broadband satellite demand for high data rates has pushed the satellite service providers towards the study of more efficient ways to exploit the available bandwidth. In this context, the new satellite systems employ multiple spot beams instead of a single global beam in the coverage area so that higher spectrum utilization is obtained by employing fractional frequency reuse among the beams [1]. Some additional benefits of multibeam systems include: (i) each beam can obtain a larger antenna gain-to-noise (G/N) ratio than in the single beam case, (ii) the available spectrum in the multibeam architecture can be reused among spatially separated beams, (iii) the multibeam architecture can support different modulations and code rates for each user depending on the user link quality (i.e. the quality of the communication link between the satellite and user terminals), (iv) multiple users can be simultaneously served in the coverage area [2].

Nevertheless, as the number of beams increases, the system performance becomes limited by the increased level of interference among the users due to the side lobes of the beams. To tackle this problem, the current multibeam systems

rely on deploying a multi-color frequency reuse pattern among the beams, i.e. adjacent beams operate in different frequency bands. Denoting N_c as the parameter which indicates the number of disjoint frequency bands, typical systems use $N_c = 4$. In another approach, the same frequency resource is duplicated within adjacent beams ($N_c = 1$). However, as mentioned earlier, interference becomes the bottleneck, and hence, application of interference mitigation techniques at the gateway is essential. To this end, interference mitigation techniques such as precoding in the forward link (i.e. the link between gateway and user terminals), and multi-user detection in the return link (i.e. the link between user terminals and gateway) have been considered in the past studies [3]-[5]. It is worth to remark that the performance of interference mitigation technique is sensitive to the quality of available Channel State Information (CSI) at the gateway. In addition, the precoding/detecting is realized at the gateway in order to guarantee a high reduction of the user terminals and payload complexity [5].

Apart from interference limitation issue, the next generation of multibeam satellite networks is expected to deliver high throughput services to small and energy efficient Mobile User Terminals (MUTs) in each of the different beams. This is referred as Multibeam Mobile Satellite Architectures (MMSA). As a matter of fact, it is common to use omni-directional low gain antennas for mobile services in MMSA. In this way, the mobility of MUTs imposes a hardware infrastructure different from the Fixed Satellite Service (FSS) with parabolic antenna. Therefore, the MUTs are usually served through the L- or S-band where the signal suffers from a lower pathloss compared to the FSS in Ka- or Ku- transmission band at the expense of a much lower available bandwidth [6].

Recently, to increase achievable rate the advanced frequency reuse among beams ($N_c = 1$) along with interference mitigation techniques have become an option in MMSA [7]. Since CSI is required to calculate interference mitigation scheme at the gateway, each MUT shall estimate the CSI of its received channel and feedback to the gateway through the satellite. Due to the nature of the MUTs, the CSI varies in time depending on the user mobility [7]. Indeed, the time varying CSI leads to outdated CSI at the gateway due to round-trip delays. Therefore, the performance of the interference mitigation scheme becomes limited compared to the case where the gateway receives real time CSI [3], [7]. So far, a few works have

addressed the application of interference mitigation techniques in MMSA. The authors in [7], [8] present the composite Rician/lognormal components of MMSA channel with correlated antennas. Further, the authors in [9] use the tools of random matrix theory to upper bound the ergodic capacity and compute the outage probability of a MMSA. Also, a multiuser decoding algorithm in return link of a MMSA is presented in [7]. The authors in [10] present design of Zero Forcing (ZF) and Minimum Mean Square Error (MMSE) in the forward link of a MMSA. The results in [7] and [10] show that the performance of interference mitigation schemes in MMSA is limited due to: (a) significant quantization error in the CSI feedback mechanism, (b) CSI delay between the gateway and MUTs. Eventually, the authors in [11] investigate employing an Adaptive Coding and Modulation (ACM) scheme in MMSA.

Contributions

The main contribution of this paper is developing precoding techniques in the forward link of a MMSA where MUTs are served in multiple beams. Since the precoding design substantially relies on how accurately the gateway knows the CSI, we investigate designing precoding schemes in the presence of delayed/outdated CSI at the gateway. To calculate the outdated CSI at the gateway, we present different CSI feedback mechanisms which aim at preserving low CSI variation and eventually provide close-to-real time CSI at the gateway. After presenting the CSI feedback mechanisms, we evaluate the throughput of closed-form linear precoding techniques by considering the ACM scheme proposed in [11] which delivers perfect rate allocation at the gateway. We also numerically show the performance of precoding schemes when either real time or outdated CSI is received at the gateway. We prove that, despite there is a performance degradation with respect to the real time CSI case, even if we have an outdated CSI, our proposed precoding schemes offer a large gain in terms of sum rate.

The rest of the paper is organized as follows. Section II presents the system model and the general assumptions of MMSA. Section III investigates the channel estimation mechanisms in MMSA. Section IV presents the interference mitigation techniques adapted with the CSI feedback mechanism in Section III. Section V shows the numerical results of the proposed interference mitigation techniques in Section IV. Finally, we draw our conclusions in Section VI.

Notation: Throughout this paper, the following notations are adopted. Boldface uppercase letters denote matrices and boldface lowercase letters refer to column vectors. $(\cdot)^H$ and $(\cdot)^T$ denote Hermitian transpose and transpose of a matrix respectively. $\mathbf{I}_N$ builds the $N \times N$ identity matrix. Boldface $\mathbf{0}$ refers to an all-zero matrix. $(\mathbf{A})_{ij}$ represents the $(i$ -th, j -th) element of matrix $\mathbf{A}$. The notation *diag* represents a diagonal matrix. Finally, $\mathbb{E}\{\cdot\}$, $\|\cdot\|$, $|\cdot|$ refer to the expected value, the Frobenius norm and absolute value operators, respectively.

II. SYSTEM DESCRIPTION

A. MMSA configuration

Consider the forward link of a MMSA, where a single Geosynchronous (GEO) satellite provides service to a set of MUTs located in a maritime area, typically operating in L-band to benefit from lower path loss. To this end, the satellite is equipped with an array fed reflector antenna comprising N feeds. The number of beams is denoted by K with $N = K$, and the MUTs are assumed to be uniformly distributed within the beams.

By employing a Time Division Multiplexing (TDM) scheme, at each time instant, a total of K single antenna MUTs, i.e. one MUT per beam, is served by a single gateway through the satellite. Figure 1 depicts a MMSA with a set of K MUTs. To focus on the precoding aspects, we consider the MUTs to satisfy two conditions:

- 1) The velocity of MUTs is not more than 70 km/h¹.
- 2) The MUT is assumed to be within a beam and handover is not considered.

Since a high throughput system is targeted, full frequency reuse among beams ($N_c = 1$) is assumed. Therefore, the inter-beam interference in the user link is the bottleneck of the whole system. Note that, the feeder link (i.e. the bidirectional link between gateway and satellite) is considered to be ideal.

B. Signal model

In the considered MMSA architecture, the received signal at any arbitrary time instant t can be modeled as

$$\mathbf{y}(t) = \mathbf{H}(t)\mathbf{x}(t) + \mathbf{n}(t), \quad (1)$$

where $\mathbf{y}(t) = [y_1(t), \dots, y_K(t)]^T$ is a $K \times 1$ vector containing the stacked received signals. The $K \times 1$ vector $\mathbf{x}(t)$ is the stacked transmitted signals at all on-board feeds. The $K \times 1$ vector $\mathbf{n}(t) = [n_1(t), \dots, n_K(t)]^T$ accounts for the zero mean Additive White Gaussian Noise (AWGN). We assume independent and identically distributed zero mean unit variance noise samples, i.e. $\mathbb{E}\{\mathbf{n}(t)\mathbf{n}(t)^H\} = \mathbf{I}_K$. In the sequel, matrix $\mathbf{H}(t)$ is the overall $K \times K$ user link channel matrix whose element $(\mathbf{H}(t))_{ij}$ represents the aggregated gain of the link between the i -th MUT and the j -th satellite feed/beam. The channel matrix can be further decomposed as [7]

$$\mathbf{H}(t) = \mathbf{D}(t)\mathbf{G}(t) \triangleq (\mathbf{h}_1^T(t), \dots, \mathbf{h}_i^T(t), \dots, \mathbf{h}_K^T(t))^T, \quad (2)$$

with:

- $\mathbf{D}(t)$ is assumed to be a $K \times K$ diagonal matrix which accounts for the attenuation in the user link with

$$\mathbf{D}(t) = \text{diag}(d_1(t), \dots, d_i(t), \dots, d_K(t)), \quad (3)$$

where $d_i(t)$ is the propagation effect which can be decomposed as $d_i(t) = d_{r,i}(t)d_{l,i}(t)$, with $d_{r,i}(t)$ and $d_{l,i}(t)$, respectively, represent complex values corresponding to the multipath rician fading due to reflections from sea surface and the classical log-normal attenuation between the satellite and i -th MUT [7].

¹The maximum speed of usual maritime vessels

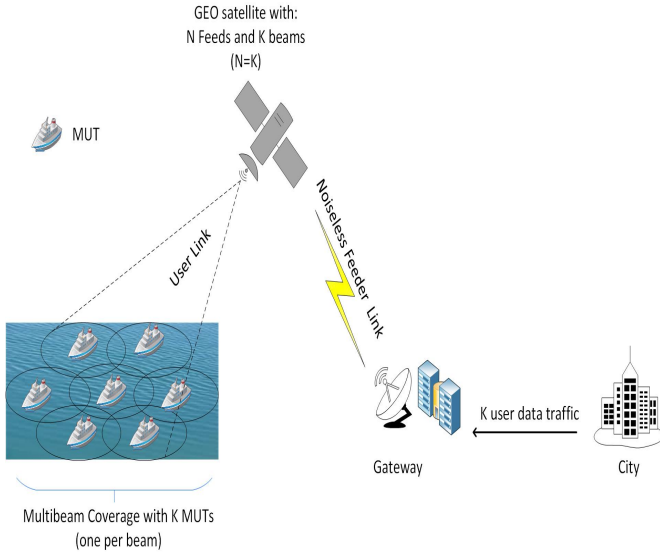


Fig. 1. Multibeam satellite system with K MUTs, one user per beam, that are uniformly distributed within coverage area.

- $\mathbf{G}(t)$ of size $K \times K$ is a square matrix that models the satellite antenna gain by $\mathbf{G}(t) = (\mathbf{g}_1^T(t), \dots, \mathbf{g}_K^T(t))^T$, where $\mathbf{g}_i(t)$ is a K dimensional row vector of $\mathbf{G}(t)$ denoting the gain for the i -th MUT (i -th row of $\mathbf{G}(t)$) with

$$\mathbf{g}_i(t) = (g_{i,1}(t), \dots, g_{i,k}(t), \dots, g_{i,K}(t)), \quad (4)$$

so that $g_{i,k}(t)$ is referring to the feed transmit gain (in power) from the k -th feed ($k = 1, \dots, N$ with $N = K$) at the satellite payload to the i -th MUT ($i = 1, \dots, K$).

- $\mathbf{h}_i(t)$ in (2) is the K dimensional row vector of $\mathbf{H}(t)$ modelling the effective channel to the i -th user (i -th row of $\mathbf{H}(t)$) and is defined as $\mathbf{h}_i(t) = d_i(t)\mathbf{g}_i(t) = d_i(t)(g_{i,1}(t), \dots, g_{i,k}(t), \dots, g_{i,K}(t))$.

We now proceed with calculating $g_{i,k}(t)$ in (4) and $d_i(t)$ in (3).

1) *Satellite Beam Gain $g_{i,k}(t)$* : The k -th satellite beam gain towards the i -th MUT, i.e. $g_{i,k}(t)$, is computed using the beam pattern [7], [12]. In particular, each MUT position within the k -th beam corresponds to an off-axis angle $\phi_{i,k}(t)$ under which the i -th MUT in beam k is illuminated. For ease of comprehension, the footprint of each beam is assumed to be a circle and that the illumination angle at the center of the beam is zero. Then, the illuminating angle $\phi_{i,k}(t)$ is computed as,

$$\phi_{i,k}(t) = \tan^{-1}\left(\frac{a_{i,k}(t)}{H}\right), \quad (5)$$

where H and $a_{i,k}(t)$, respectively, represent the height of the satellite, and the distance of the i -th MUT located in the k -th beam to the center of beam k . Employing this illuminating angle, we can obtain the satellite beam gain as [12],

$$g_{i,k}^2(t) = \left(\frac{(\rho+1)(1+\chi)}{(\rho+1)(1+\chi)+\chi} \Omega(t) \right)^2, \quad (6)$$

with

$$\Omega(t) \triangleq \left(\frac{2J_1(u(t))}{u(t)} + 2^{\rho+1} \rho! \frac{\chi}{1-\chi} \frac{J_{\rho+1}(u(t))}{u^{\rho+1}(t)} \right), \quad (7)$$

and $u(t) = \lambda^{-1} \pi d_a \sin(\phi_{i,k}(t))$. The notation d_a denotes the aperture diameter of the transmitting antenna, χ is the aperture edge taper, and λ is the wavelength. In addition, the operator $J_\rho(u(t))$ denotes the Bessel function of the first kind with order ρ , where ρ is a positive integer.

2) *Attenuation $d_i(t)$* : The radio propagation characteristics, i.e. $d_i(t)$ in (3), for the considered scenario can be modelled as [13],

$$d_{r,i}(t) = \left(\sqrt{\frac{L_i}{L_i+1}} d_{r,i}^{LOS} e^{j2\pi f Z_{i,LOS}(t)} + \sqrt{\frac{1}{L_i+1}} d_{r,i}^{NLOS} e^{j2\pi f Z_{i,NLOS}(t)} \right) \quad (8)$$

$$d_{l,i}(t) = \sqrt{d_{\text{shadow},i}(t)}, \quad (9)$$

where L_i is the Rician factor. The parameters $e^{j2\pi f Z_{i,LOS}(t)}$ and $e^{j2\pi f Z_{i,NLOS}(t)}$ will be defined later in this subsection. The term $d_{r,i}^{LOS}(t)$ models the Line of Sight (LOS) deterministic component with $0 < d_{r,i}^{LOS}(t) < 1$. The term $d_{r,i}^{NLOS}$ is a Non Line of Sight (NLOS) complex random component representing the scattered component dominated by the reflections from the sea-surface and is normally distributed with zero mean and variance σ_{NLOS}^2 . The Rician factor L_i determines the multipath strength. Furthermore, $d_{\text{shadow},i}(t)$ represents the impact of shadowing at the i -th MUT and its Probability Density Function (PDF) is

$$f(d_{\text{shadow},i}(t)) = \frac{1}{d_{\text{shadow},i}(t) \sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(\ln(d_{\text{shadow},i}(t)) - \mu_i)^2}{2\sigma_i^2}\right), \quad (10)$$

where $d_{\text{shadow},i}(t) > 0$. In (10), μ_i and σ_i^2 are the mean and variance of $d_{\text{shadow},i}(t)$, respectively. Indeed, the value of $d_{l,i}(t)$ in the maritime area refers to the presence of clouds on the top of the i -th beam.

It is necessary to point out that $d_{r,i}(t)$ in (8) and $d_{l,i}(t)$ in (9) vary in time depending on each MUT speed and the environment that MUT is located [7]. Moreover, the mobility of the MUT causes the Doppler effect. The Doppler effect on the received signals at i -th MUT for LOS and NLOS components are respectively denoted by $e^{j2\pi f Z_{i,LOS}(t)}$ and $e^{j2\pi f Z_{i,NLOS}(t)}$ in (8). The Doppler effect on the received signals at MUTs affects on the channel Coherence Time at i -th MUT. In this study, the Doppler effect is modeled using a Butterworth filter [14] with a frequency response of order κ_f , as,

$$|B(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_c}\right)^{2\kappa_f}}, \quad (11)$$

where f_c and f are the cut-off frequency and carrier frequency, respectively. For the MUT with speed ν m/s, the D_{LOS} and D_{NLOS} in (8) can be computed by [13],

$$Z_{i,LOS}(t) = \frac{\nu}{\lambda} \cos(\phi_{i,k}(t))$$

$$Z_{i,NLOS}(t) = \frac{\nu}{\lambda} \cos(\phi_{i,k}^{NLOS}(t)), \quad (12)$$

with $\phi_{i,k}^{LOS}(t)$ and $\phi_{i,k}^{NLOS}(t)$ respectively denote the illumination angle that the i -th MUT receives the LOS and NLOS components as formulated in subsection II.A.1. Considering that the maritime area provides the degree of freedom of $\cos(\phi_{i,k}^{LOS}(t)) \approx \cos(\phi_{i,k}^{NLOS}(t))$ [15], we have $Z_i(t) \triangleq Z_{i,LOS}(t) \approx Z_{i,NLOS}(t)$. Then, the Coherence Time according to Clarke's model can be computed using,

$$\vartheta_{CT_i} = \frac{9}{16\pi Z_i(t)}. \quad (13)$$

In this way, $d_{r,i}(t)$ in (8) can be eventually rewritten as

$$d_{r,i}(t) = \left(\sqrt{\frac{L_i}{L_i + 1}} d_{r,i}^{LOS} \exp^{j2\pi f Z_i(t)} + \sqrt{\frac{1}{L_i + 1}} d_{r,i}^{NLOS} \exp^{j2\pi f Z_i(t)} \right). \quad (14)$$

C. Gateway processing

Let us consider that the gateway employs a precoding scheme to mitigate the inter-beam interference. For any time instant t , the incorporation of the precoder in (1) results in,

$$\mathbf{x}(t) = \mathbf{W}(t)\mathbf{u}(t), \quad (15)$$

where $\mathbf{u}(t) = (u_1(t), \dots, u_i(t), \dots, u_K(t))^T$ is the transmit symbol vector of size $K \times 1$ such that the i -th entry of $\mathbf{u}(t)$, i.e. $u_i(t)$, is the symbol destined to the i -th MUT with $E\{\mathbf{u}(t)\mathbf{u}^H(t)\} = \mathbf{I}_K$. The matrix $\mathbf{W}(t) \triangleq \beta(\mathbf{w}_1(t), \dots, \mathbf{w}_i(t), \dots, \mathbf{w}_K(t))$ denotes an $K \times K$ precoding matrix and $\mathbf{w}_i(t)$ of size $K \times 1$ is the i -th column of the matrix $\mathbf{W}(t)$. Moreover, the value of constant β has to be chosen to fulfill either the following total power constraints depending on the nature of operations,

$$\beta = \frac{P}{\text{trace}(\mathbf{W}(t)\mathbf{W}^H(t))}, \quad (16)$$

or following per-antenna/feed power constraint,

$$\beta = \frac{P}{K \text{trace}(\mathbf{W}(t)\mathbf{W}^H(t))_{kk}} \quad k = 1, \dots, K \quad (17)$$

with P denoting the total transmit power at the satellite.

The overall sum-rate for the i -th MUT is denoted by $SR_i(t)$ and is defined by,

$$SR_i(t) = \log_2(1 + \text{SINR}_i(t)), \quad (18)$$

where $\text{SINR}_i(t)$ denotes the received Signal to Interference plus Noise Ratio (SINR) at the i -th MUT and is expressed as,

$$\text{SINR}_i(t) = \frac{|\mathbf{h}_i^H(t)\mathbf{w}_i(t)|^2}{\sum_{j \neq i} |\mathbf{h}_i^H(t)\mathbf{w}_j(t)|^2 + 1}. \quad (19)$$

As stated in Section I, the performance of precoding technique $\mathbf{W}(t)$ is sensitive to the quality of CSI at the gateway. To this end, the MUTs shall estimate the CSI and feedback to the gateway through the satellite. In addition, due to the mobility of MUTs, the CSI is time-varying. With this, the

instantaneous CSI used for the precoding (as described above) can be outdated due to CSI variation in consecutive time instants along with the latency introduced by the round trip travel (fractions of a second) and processing. Therefore, a proper time-varying channel estimation mechanism needs to be employed in the presence of the precoding.

Remark 1. In general, the CSI estimation delay appears due to movements of either MUTs or related to payload imperfections (phase noise) and satellite movements [11]. We consider that the CSI delay is only due to the mobility of MUTs at the multibeam coverage area and the satellite is perfectly calibrated to the gateway through a geostationary orbit. Moreover, the feedback mechanism is assumed to be ideal, with no packet or data loss.

The next section aims to provide low complex but reliable channel estimation mechanisms.

III. CHANNEL ESTIMATION MECHANISM

To determine uncertainty of CSI due to the mobility of MUTs, a number of past efforts generally in terrestrial networks have addressed the channel estimation approaches [11]-[16]. Mathematically, the main contribution of CSI estimation mechanism is to determine the variation of $\mathbf{D}(t)$ and $\mathbf{G}(t)$ in (2) at the gateway. In this context, a low complex channel estimation approach is to consider the outdated information of $\mathbf{D}(t)$ and $\mathbf{G}(t)$ at the gateway [11]-[16]. Then, the precoding matrix $\mathbf{W}(t)$ can be constructed through outdated information of $\mathbf{D}(t)$ and $\mathbf{G}(t)$. In such a scenario, $\mathbf{D}(t-\tau)$ and $\mathbf{G}(t-\tau)$ represent respectively the outdated information of $\mathbf{D}(t)$ and $\mathbf{G}(t)$ where τ denotes the period between the time interval that an MUT estimates its CSI, i.e. the i -th MUT estimates $(d_i(t)\mathbf{g}_i(t))$, and the time instant that the precoding matrix is calculated based on this CSI at the gateway. The period τ consists of the following contributions [10]:

- (a) CSI feedback to the gateway is 250 ms.
- (b) The time taken by the gateway to calculate the precoding matrix is assumed to be 100 ms.
- (c) The propagation time to transmit the signals from the gateway to the MUTs is 250 ms.
- (d) The user terminal processing time is assumed to be 100 ms.

The estimates in (b) and (d) above vary with the architecture of the gateway and MUTs. The authors in [10] have expressed an approximation of $\tau = 500$ ms, i.e. the impact of (a) and (c). In this work, we also consider $\tau = 500$ ms and more precise calculation of τ is left for the future works².

In any time instant t , the outdated CSI of $\mathbf{D}(t-\tau)$ and $\mathbf{G}(t-\tau)$ at the gateway can be collected in two ways:

Memoryless Scheme (MLS), refers to the CSI estimation with a delay equal to 500 ms, i.e. $\tau = 500$ ms. Throughout this paper $\mathbf{D}(t-\tau)$ and $\mathbf{G}(t-\tau)$ refer the estimation of $\mathbf{D}(t)$ and $\mathbf{G}(t)$ with a delay of $\tau = 500$ ms.

²Indeed, the effects of (b) and (d) can be changed according to the considered MMSA. On the contrary, the impacts of (a) and (c) appear in all MMSA systems [10].

Memory based Scheme (MS), refers to the CSI estimation during a set of time instance other than t ; in particular, the CSI acquired at $\tau = \tau_i$, with $i = 1, \dots$ will be used. Obviously, to implement MS, the gateway should be equipped with a memory unit aiming to compute CSI in a set of consecutive time intervals τ_i . Note that the delay between consecutive moments of τ_i is considered to be 500 ms. For instance, the delay between the $i = l - 1$ and $i = l - 2$ or l is 500 ms. Throughout this work the matrices $\mathbf{D}(t - \tau_i)$ and $\mathbf{G}(t - \tau_i)$ represent the estimation of $\mathbf{D}(t)$ and $\mathbf{G}(t)$ in a set of $\{\tau_i\}_i$ previous time series than t .

By employing the proposed MS and MLS approaches, the following CSI estimation strategies can be conceived:

(i) **MLS with Fixed Beam Gain (MLS-FBG)**: Herein, the design of precoding in MLS is targeted where the outdated $\mathbf{D}(t - \tau)$ is available at the gateway. In addition, it is assumed that the beam gain, $\mathbf{G}$, is fixed, i.e.,

$$\mathbf{G} \triangleq \mathbf{G}(t - \tau) = \mathbf{G}(t), \quad (20)$$

and is known at the gateway. The equality in (20) implies that each MUT receives a fixed gain from $\mathbf{G}$ even if it moves from one point to another point within the beam. This assumption can be motivated from the fact that the angle separation among different points of the same beam is considered to be very small. Moreover, the slant path variations due to the mobility of the MUTs, can be considered negligible due to the very long Earth-Space distance. With assuming the perfect information of $\mathbf{G}$, the main challenge of designing precoding in MLS-FBG is to deal with the outdated information of attenuation matrix $\mathbf{D}(t - \tau)$.

It is also worth to remark here that determining information of $\mathbf{D}(t - \tau)$ and $\mathbf{G}$ separately at the gateway presents a degree of freedom where the precoding scheme can be designed in order to initially mitigate inter-beam interference in $\mathbf{G}$ and then carefully equalize the impact of $\mathbf{D}(t - \tau)$ to improve the achievable rate at each MUT. To calculate $\mathbf{D}(t - \tau)$ at the gateway, two approaches can be realized:

(i.1.) *MUT oriented estimation*: It is considered that the MUTs, in addition, to the gateway know the fixed information of $\mathbf{G}$. Then, the MUTs split and feedback individual information of $\mathbf{D}(t - \tau)$ to the gateway.

(i.2.) *Gateway oriented estimation*: MUTs report $\mathbf{H}(t - \tau)$ to the gateway, and the gateway finds $\mathbf{D}(t - \tau)$ from $\mathbf{D}(t - \tau) = \mathbf{H}(t - \tau)\mathbf{G}^{-1}$.

(ii) **MS with Fixed Beam Gain (MS-FBG)**: Herein, the design of precoding in MS is targeted where the same setting of MLS-FBG is realized considering $\mathbf{D}(t - \tau_i)$ with τ_i , $i = 1, \dots, t - 1$, time intervals. Fixed $\mathbf{G}$ in (20) is considered at the gateway. Further, $\mathbf{D}(t - \tau_i)$ can be estimated using $\mu(t)$ (say) measurements,

$$\mathbf{D}_{avg}(t) \triangleq \frac{\sum_{i=1}^{\mu(t)} \mathbf{D}(t - \tau_i)}{\mu(t)}. \quad (21)$$

In fact, $\mathbf{D}_{avg}(t)$ represents the average of $\mathbf{D}(t - \tau_i)$ for $\mu(t)$ measurements.

With employing aforementioned approaches in MLS-FBG and MS-FBG, the following section provides designing different precoding techniques that establish a competent trade off between complexity and performance.

IV. INTERFERENCE MITIGATION TECHNIQUES

In this section, we investigate novel precoding schemes in the following possible scenarios:

- **Scenario 1**: The goal is to develop MMSE and ZF precoding through employing the channel estimation mechanism described in MLS-FBG.
- **Scenario 2**: This scenario provides a novel precoding technique via using channel estimation mechanism in MS-FBG, namely: two stage precoding.

We now proceed to formulate the precoding design in the above mentioned scenarios.

A. Precoding design in Scenario 1

At any time instant t , let's consider the gateway knows the information of $\mathbf{D}(t - \tau)$ and fixed $\mathbf{G}$, i.e. $\mathbf{G}(t) = \mathbf{G}(t - \tau)$. The Mean Square Error (MSE) for the i -th MUT in (1) can be formulated by $\text{MSE}_i(t) = \mathbb{E}\{|y_i(t) - u_i(t)|^2\}$. In fact, $\text{MSE}_i(t)$ is the conditional MSE, conditioned on time varying channel gains. It is also well known that the mathematical expression of Sum MSE (SMSE) for a set of K MUTs becomes,

$$\begin{aligned} \text{SMSE}(t) &= \mathbb{E}\{\|\mathbf{y}(t) - \mathbf{u}(t)\|^2\} \\ &= \mathbb{E}\{\|(\mathbf{D}(t)\mathbf{G}\mathbf{W}(t)\mathbf{u}(t) + \mathbf{n}(t)) - \mathbf{u}(t)\|^2\}. \end{aligned} \quad (22)$$

Then, the SMSE performance metric, neglecting the precoder-independent noise variance, can be simplified as,

$$\text{SMSE}(t) = \text{trace}[(\mathbf{D}(t)\mathbf{G}\mathbf{W}(t) - \mathbf{I}_K)(\mathbf{W}^H(t)\mathbf{G}^H\mathbf{D}^H(t) - \mathbf{I}_K)]. \quad (23)$$

The problem of optimizing SMSE can be then formulated as,

$$\begin{aligned} \min_{\mathbf{W}(t)} \quad & \text{trace}[(\mathbf{D}(t)\mathbf{G}\mathbf{W}(t) - \mathbf{I}_K)(\mathbf{W}^H(t)\mathbf{G}^H\mathbf{D}^H(t) - \mathbf{I}_K)] \\ \text{s.t.} \quad & \text{trace}(\mathbf{W}(t)\mathbf{W}^H(t)) \leq P. \end{aligned} \quad (24)$$

Motivated by (24), in this scenario, the ZF and MMSE precoders are derived using information about $\mathbf{G}$ in (20) and $\mathbf{D}(t - \tau)$ as,

$$\begin{aligned} \mathbf{W}_{\text{ZF}}(t) &= \beta \hat{\mathbf{H}}^H(t)(\hat{\mathbf{H}}(t)\hat{\mathbf{H}}^H(t))^{-1}, \\ \mathbf{W}_{\text{MMSE}}(t) &= \beta \hat{\mathbf{H}}^H(t)(\hat{\mathbf{H}}(t)\hat{\mathbf{H}}^H(t) + \mathbf{I}_K)^{-1}, \quad \text{where} \\ \hat{\mathbf{H}}(t) &= \mathbf{D}(t - \tau)\mathbf{G} \end{aligned} \quad (25)$$

B. Precoding design in Scenario 2

Consider now the gateway exploits fixed $\mathbf{G}$, and outdated $\mathbf{D}(t - \tau_i)$, for $i = t - 1, t, \dots$ to obtain the precoder as,

$$\mathbf{W}(t) = \beta \mathbf{W}_a(t)\mathbf{W}_b(t) \quad \text{where} \quad (27)$$

$$\mathbf{W}_a(t) = \beta \mathbf{H}_{avg}^H(t)(\mathbf{H}_{avg}(t)\mathbf{H}_{avg}^H(t))^{-1}, \quad (28)$$

$$\mathbf{H}_{avg}(t) = \mathbf{D}_{avg}(t)\mathbf{G}$$

where $\mathbf{W}_a(t)$ of size $K \times K$ is used to tackle with the inter-beam interference.

Then, $\mathbf{W}_b(t)$ of size $K \times K$ is a diagonal matrix which optimizes the minimum achievable rate obtained at MUTs. In this way, the noise-free SMSE can be calculated as,

$$\text{SMSE}(t) = \text{trace}[\mathbf{I}_K - 2\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\} + \mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\mathbf{W}_b^H(t)(\mathbf{D}_{avg}^H(t))^{-1}\mathbf{D}^H(t)].$$

Then, the underlying optimization problem can be formulated as,

$$\min_{\mathbf{W}(t)} \text{trace}[\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\mathbf{W}_b^H(t)(\mathbf{D}_{avg}^H(t))^{-1}\mathbf{D}^H(t) + \mathbf{I}_K - 2\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\}] \quad (29)$$

$$\text{s.t.} \quad \mathbf{W}(t) = \beta \mathbf{W}_a(t)\mathbf{W}_b(t) \quad (30)$$

$$\text{trace}(\mathbf{W}^H(t)\mathbf{W}(t)) \leq P \quad (31)$$

Clearly, the objective function in (29) is equivalent to,

$$\begin{aligned} & \text{trace}[\mathbf{D}(t)\mathbf{D}_{avg}(t)^{-1}\mathbf{W}_b(t)\mathbf{W}_b^H(t)(\mathbf{D}_{avg}^H(t))^{-1}\mathbf{D}^H(t) \\ & - 2\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\} + \mathbf{I}_K] = \\ & \sum_{i=1}^K 1 + \lambda_i \left(\mathbf{W}_b(t)^H (\mathbf{D}_{avg}(t)^H)^{-1} \mathbf{D}^H(t) \mathbf{D}(t) \mathbf{D}_{avg}^{-1}(t) \mathbf{W}_b(t) \right. \\ & \left. - 2\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\} \right), \end{aligned} \quad (32)$$

where $\lambda_i(\cdot)$ denotes the i -th eigenvalue of the respective matrix. Since $\mathbf{W}_b(t)$ is diagonal, it can be denoted as $\mathbf{W}_b(t) \triangleq \text{diag}(\sigma_1(t), \dots, \sigma_i(t), \dots, \sigma_K(t))$. Without loss of generality, we assume σ_i to be real.

Due to the diagonal nature of the matrices involved, it immediately follows that $\lambda_i \left(\mathbf{W}_b(t)^H (\mathbf{D}_{avg}(t)^H)^{-1} \mathbf{D}^H(t) \mathbf{D}(t) \mathbf{D}_{avg}^{-1}(t) \mathbf{W}_b(t) \right) = \sigma_i^2(t) \rho_i(t)$ and $\lambda_i \left(\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\mathbf{W}_b(t)\} \right) = \sigma_i(t) \delta_i(t)$, where, $\rho_i(t) \triangleq \lambda_i \left((\mathbf{D}_{avg}^H(t))^{-1} \mathbf{D}^H(t) \mathbf{D}(t) \mathbf{D}_{avg}^{-1}(t) \right)$ and $\delta_i(t) \triangleq \lambda_i \left(\text{Re}\{\mathbf{D}(t)\mathbf{D}_{avg}^{-1}(t)\} \right)$.

With these notations, it suffices to find the entries $\{\sigma_i(t)\}$; the optimal $\sigma_i(t)$ is a direct consequence of the following Proposition.

Proposition 1. *The optimal $\sigma_i(t)$ can be obtained by solving the following convex problem [17],*

$$\min_{\sigma_i(t)} 1 + \sigma_i^2(t) \rho_i(t) - 2\delta_i(t) \sigma_i(t), \quad i \in [1, K] \quad (33)$$

Solution of the problem in (33) is trivially given by

$$\frac{\partial(1 + \sigma_i(t)\lambda(t)\sigma_i(t) - 2\lambda_{im}(t)\sigma_i(t))}{\partial\sigma_i(t)} = 0 \Rightarrow$$

$$\sigma_i(t) = \frac{\delta_i(t)}{\rho(t)}. \quad (34)$$

For the proof, see Proposition 1 in [17]. ■

It is worth to remark that to design $\mathbf{W}_b(t)$ above, instantaneous $\mathbf{D}(t)$ is required to calculate $\rho_i(t)$ and $\delta_i(t)$. Since the instantaneous $\mathbf{D}(t)$ is not available at the gateway, we propose

to find $\mathbf{W}_b(t)$ using $\mathbf{D}(t - \tau)$ instead of $\mathbf{D}(t)$. Finally, the power constraint is satisfied by the appropriate chose of the scaling parameter β .

V. NUMERICAL RESULTS

In order to compare the performance of the proposed precoding schemes in Scenarios 1 and 2, Monte Carlo simulations have been carried out in this section. The simulation setup is based on an array fed reflector antenna/feed provided by European Space Agency (ESA) in the context of SatNex IV project [18]. The number of feeds and beams are assumed to be $N = K = 16$ and are assumed to illuminate the maritime coverage area. We assume K MUTs, one MUT per beam, are served in the coverage area. The rest of the simulation parameters are collected in Table I. Further $\mathbf{D}_{avg}(t)$ is obtained from averaging $\mu(t) = 500$ realizations.

Results are averaged for a total of 400 user link channel realizations. The channel has the following characteristics:

- The Doppler effect and the Coherence Time of the channel components in different speed can be obtained by (13). For instance, the maximum Doppler effect and the Coherence Time for a carrier frequency of 1.59 GHz and speed of 21.6 km/h are 32 Hz and 0.0313 sec, respectively.
- The randomness of the channel is due to the mobility of MUTs which are assumed to be uniformly distributed within the beams, i.e., one MUT per beam with maximum speed 70 Km/h (maximum possible speed of vessels).
- Each user employs all available bandwidth of beam it is located in.

TABLE I
USER LINK SIMULATION PARAMETERS [18]

Parameter	Value
Coverage	Maritime area
Satellite height	35786 km (GEO)
Earth radius	6378.137 Km
Number of feeds N	16
Speed of MUT	1 to 70 Km/h
Number of beams K	16
User location distribution	Uniformly distributed (one MUT per beam)
Carrier frequency	1.59 GHz (L band)
Total bandwidth	800 KHz
Receiver noise temperature	235
User antenna gain	-13.5 dB/K
Carrier over Multipath ratio	15 dB
Feed radiation pattern	Provided by ESA [18]

A. Performance metrics

The figures of merit are presented in this section. We obtain the throughput of the precoding schemes proposed in the context of Scenarios 1, 2 versus different speed of MUTs. For this, we evaluate the throughput of the aforementioned precoding schemes via a subset of the modulation and coding (MODCOD) options that ETSI TS 102 744-2-1 standard communication [19] offers. Note that the standard is

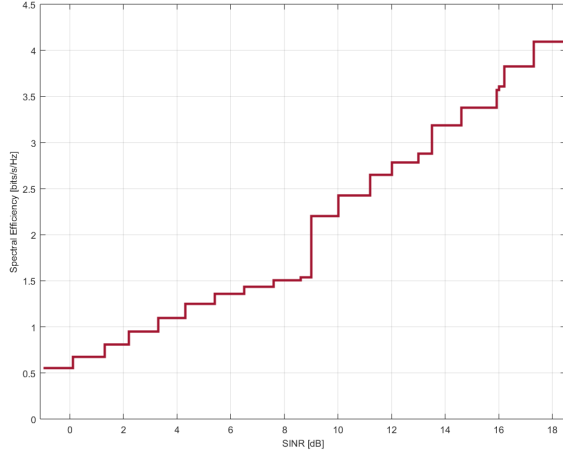


Fig. 2. ETSI TS 102 744-2-1 MODCOD subset.

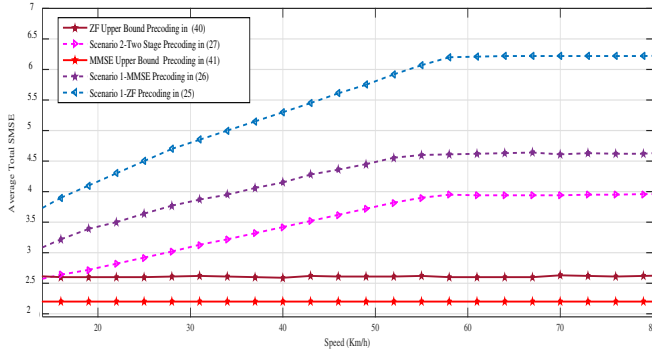


Fig. 3. Average total SMSE resulted from precoding schemes in Scenario 1 and 2.

used as a reference system for mobile interactive narrowband communications. Figure 2 depicts the MODCODs subsets of ETSI TS 102 744-2-1 with a target packet error rate of 10^{-3} .

For a best practice, at arbitrary time instant t , we also consider two upper bound reference scenarios as,

$$\mathbf{W}_{\text{ZF, upper}}(t) = \beta \mathbf{H}(t)^H (\mathbf{H}(t) \mathbf{H}(t)^H)^{-1}, \quad (35)$$

$$\mathbf{W}_{\text{MMSE, upper}}(t) = \beta \mathbf{H}(t)^H (\mathbf{H}(t) \mathbf{H}(t)^H + \mathbf{I}_K)^{-1} \quad (36)$$

where $\mathbf{W}_{\text{ZF, upper}}(t)$ and $\mathbf{W}_{\text{MMSE, upper}}(t)$ refer to design of ZF and MMSE schemes when the real time information of channel matrix $\mathbf{H}(t)$ in (1) is available at the gateway.

B. Results

Fig. 3 depicts the evolution of the total average SMSE as a function of MUTs speed for different scenarios. From Fig. 3, it is seen that:

- a The upper bound reference scenarios in (35) and (36) provide stable state for any variation of MUTs speed. The fact is that the proposed upper bound references

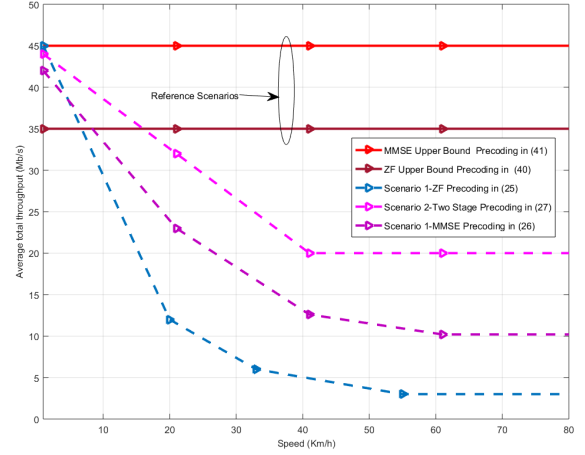


Fig. 4. Average throughput comparison of precoding schemes in Scenario 1 and 2 based on ETSI TS 102 744-2-1 MODCODs in Fig. 2

are perfectly exploiting the channel condition since the real time (perfect) CSI is available at the gateway.

- b Two stage precoding in (27) outperforms the other precoding techniques and presents results closer to the upper bound schemes in (35) and (36). Thus designing precoding technique with a set of $\{\tau_i\}_i$ outdated CSI in MS mode performs better than designing similar schemes with employing the outdated CSI at τ with 500 ms delay in MLS mode. Further, the precoders in (27) are robust to channel variations in comparison with other techniques.
- c The performance of proposed precoding schemes except upper bound reference in (35) and (36) is lowered by increasing the speed. This comes from the fact that, even if the precoders perfectly mitigate the impact of $\mathbf{G}(t)$, the precoding techniques are not able to properly equalize the attenuation effect $\mathbf{D}(t)$ in high speeds. This is due to decreasing the correlation between the available outdated CSI of $\mathbf{D}(t)$ at the gateway and the real time attenuation components $\mathbf{D}(t)$ in the channel ; both these become uncorrelated at speeds in excess of 50 ~ 55 Km/h.

The improvement can be also noticed from the corresponding average total throughput plot (Mbits/s) versus the speed of MUTs in Fig. 4. The expected result of SMSE in Figure 3 is corroborated by the throughput plot in Fig. 4 which is established under MODCOD allocation. In other words, two stage precoding in (27) presents results closer to the upper bound in (35) and (36) than other scenarios.

VI. CONCLUSION

This paper evaluated the design of precoding in Multibeam Mobile Satellite Architectures (MMSA). In contrast to the multibeam FSS, an additional challenge comes into picture in the MMSA with MUTs in the coverage area: a time varying CSI. The performance sensitivity of the precoding

schemes to the quality of CSI at the gateway, and the availability of only the outdated CSI at the gateway tends to limit the precoding gains. Towards this, the paper develops CSI feedback mechanisms aimed at mitigating the impact of outdated CSI. In particular, the paper has developed two CSI feedback mechanisms, one based only on the latest CSI estimation, i.e., MLS approach, and the other based on all the previous CSI estimations, i.e., the MS approach. Finally, the paper exploits the developed techniques to construct different precoding schemes which are adapted with the proposed CSI feedback mechanism.

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